PYTHON DICTIONARY

DICTIONARY

- It is an unordered set of key and value pair. The pair (key and value) is called as item.
- The key and value are separated by a **colon (:) operator**. Items are separated by comma (,) operator
- It is a collection which is unordered, changeable and indexed.
- Unlike set, it is a two end data structure
- Like list and set, it is used to store homogenous and heterogeneous items
- Unlike list and set, it maintains key and value pairs
- Each value stored in a dictionary can be accessed using a key, which is any type of object (Ex. int, string, list, etc, …) instead of using its index to address it.
- Like list, it supports indexing
- Here keys are unique and value includes duplicate elements.

MAIN OPERATIONS

- Storage and Retrieval
- Searching and Deletion
- Updation

DIFFERENCE BETWEEN SET AND DICTIONARY

S.N	Set	Dictionary
1.	It is a one end data structure	It is two end data structures like key and value
2.	It does not accept duplicate values	In dictionary, **key should be unique** but value supports duplicate elements
3.	It does not support indexing	It supports indexing

DICTIONARY CREATION

- In python, the dictionary is created using two ways. They are
  1. Using **curly braces {} symbol**
  2. Using **built-in function dict()**
- The elements in the dictionary are separated by comma operator

Method 1 (using curly braces)

Syntax

```
dictionary-name={k1:v1, k2:v2, …kn:vn}
```

Example (dictionary with key-value pair elements)

```
db={5:”Salem”,7:”Chennai”,4:”Madurai”,1:”Trichy”}
```

Example (empty dictionary)

```
db={}
```

Method 2 (using built-in dict() function)

Syntax

```
dictionary-name=dict({list of values})
```

Example (dictionary with key-value pair elements)

```
db= dict({5:"Salem",7:"Chennai",4:"Madurai",1:"Trichy"})
```

Example (empty dictionary)

```
db=dict({})
```

ACCESSING ELEMENTS IN DICTIONARY

- We can access the individual item or items of a dictionary by using key name
- The key name can be specified inside the square brackets or using built-in method called **get(key)**

Example (Accessing particular item)

```
dt={1:"salem",3:"chennai",6:"madurai"}
ele=dt[3]
```

 # using indexing

OR

Example (Accessing particular item)

```
dt={1:"salem",3:"chennai",6:"madurai"}
ele=dt.get(3)
```

 # using get(key)

COMMON BUILT-IN METHODS OF DICTIONARY

1. len(dictionary name)
   - It is used to return the length of a dictionary (total number of elements in a dictionary)
   - It takes only one argument. The argument is dictionary name
   - Return type: int
2. keys()
   - It is used to list **all the key elements of dictionary**
   - It does not take any arguments
   - Return type: dict_key (int, string, list, etc, …)
3. values()
   - It is used to list **all the value elements of dictionary**
   - It does not take any arguments
   - Return type: dict_value (int, string, list, etc, …)

4. items()
   - It is used to list **all the key and value elements of dictionary**
   - It does not take any arguments
   - Return type: dict_items
5. get(key)
   - It is used to **display / return the value of the given key**
   - It takes only one argument. The argument is key
   - Return type: value type (int, string, list, etc, …)
6. clear()
   - It is used to delete all the items in a dictionary
   - It does not take any arguments
   - Return type: void
7. copy()
   - It is used to copy the contents of dictionary
   - It does not take any arguments
   - Return type: dictionary
8. pop(key, value)
   - It is used to remove the item from dictionary based on the key
   - It takes two arguments. The first argument is key and second argument is value. It is an optional
   - Return type: value type (int, str, list, etc, …)
9. del dictionary or del dictionary[key]
   - It is used to remove one or more items from the dictionary
   - It takes only one argument. The argument is either dictionary-name or dictionary-name[key]
   - del dictionary-name
     - It is used to delete entire dictionary
   - del dictionary-name[key]

- It is used to delete the particular item based on the key

- Return type: void

I. CREATION OF PYTHON DICTIONARY

(Storage and Retrieval)

(welcome.py)

Language : Python 3.7 (Python 3)

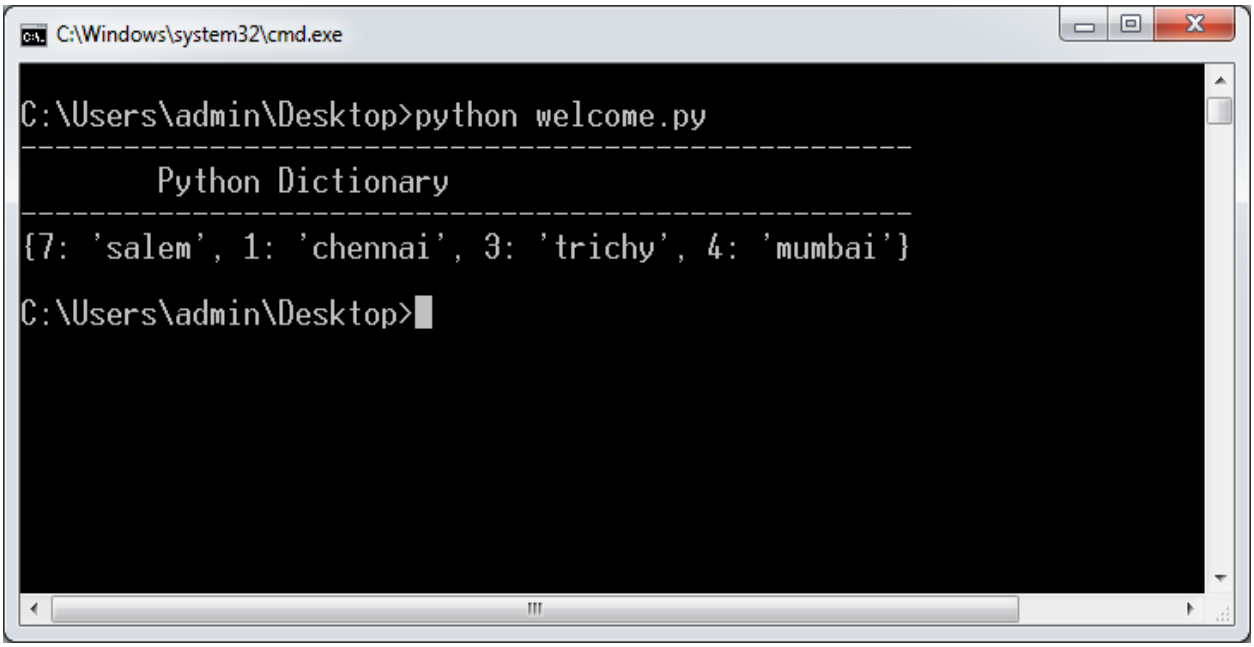
Application Type : Console App

Python Dictionary Creation

1. SOURCE CODE

```
print("-----------------------------------------------")
print("\tPython Dictionary")
print("-----------------------------------------------")
# dictionary creation
db={7:"salem",1:"chennai",3:"trichy",4:"mumbai"}
# print elements of dictionary
print(db)
```

2. OUTPUT



II. DICTIONARY RETRIEVAL

(Storage, and Retrieval)

(welcome.py)

Language : Python 3.7 (Python 3)

Application Type : Console App

1. SOURCE CODE

```
print("-----------------------------------------------")
print("\tPython Retrieval")
print("-----------------------------------------------")
# dictionary creation
db={7:"salem",1:"chennai",3:"trichy",4:"mumbai"}
# print contents of dictionary using for loop
print("Contents of Python Dictionary:")
for i,j in db.items():
        print(i,"->",j)
# show keys of dictionary using for loop
print("-----------------------------------------------")
print("Keys of Dictionary:")
for k in db.keys():
        print(k)
print("-----------------------------------------------")
# show all values of dictionary using for loop
print("Values of Dictionary:")
for v in db.values():
        print(v)
print("-----------------------------------------------")
# show the total length of dictionary
print("Total Elements of Dictionary: ",len(db))
```

Python Dictionary Creation

Retrieval of dictionary keys and values

Retrieval of dictionary keys

Retrieval of dictionary values

2. OUTPUT

```
C:\Windows\system32\cmd.exe
C:\Users\admin\Desktop>python welcome.py
-----------------------------------------------
        Python Retrieval
-----------------------------------------------
Contents of Python Dictionary:
7 -> salem
1 -> chennai
3 -> trichy
4 -> mumbai
-----------------------------------------------
Keys of Dictionary:
7
1
3
4
-----------------------------------------------
Values of Dictionary:
salem
chennai
trichy
mumbai
-----------------------------------------------
Total Elements of Dictionary:  4

C:\Users\admin\Desktop>
```

III. DICTIONARY INDEX BASED INSERTION

(Storage, and Retrieval)

(welcome.py)

Language : Python 3.7 (Python 3)

Application Type : Console App

1. SOURCE CODE

```
print("-----------------------------------------------------------------")
print("\t\tPython Insertion using Indexing")
print("-----------------------------------------------------------------")
# empty dictionary creation
db={}
# add elements using index position
db[5]="CTS"
db[4]="TCS"
db[2]="Microsoft"
db[1]="Google"
# show elements using object name
print(db)
print("-----------------------------------------------------------------")
```

Python Dictionary Creation

2. OUTPUT

```
C:\Windows\system32\cmd.exe

C:\Users\admin\Desktop>python welcome.py
-----------------------------------------------------------------
                Python Insertion using Indexing
-----------------------------------------------------------------
{5: 'CTS', 4: 'TCS', 2: 'Microsoft', 1: 'Google'}
-----------------------------------------------------------------

C:\Users\admin\Desktop>
```

IV. DICTIONARY DELETION

(Storage, Deletion and Retrieval)

(welcome.py)

Language : Python 3.7 (Python 3)

Application Type : Console App

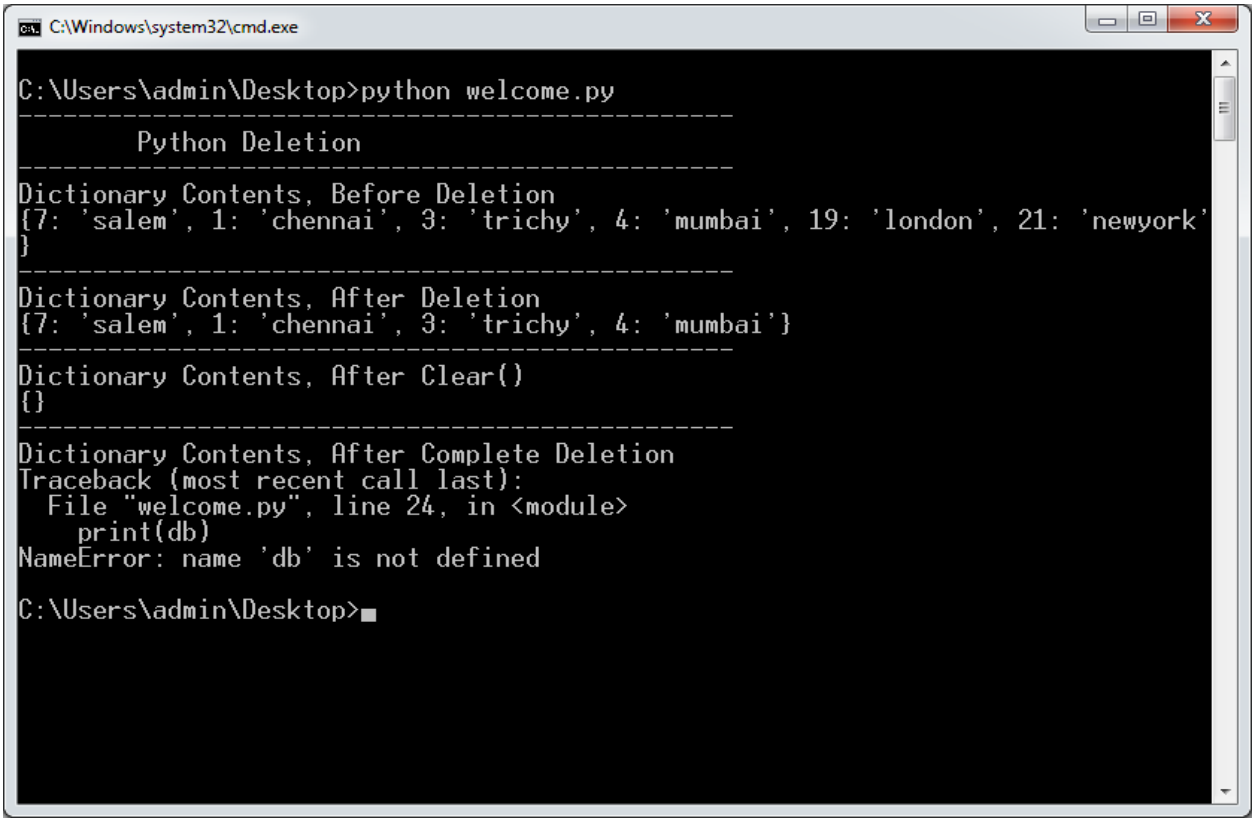
1. SOURCE CODE

```
print("-----------------------------------------------")
print("\tPython Deletion")
print("-----------------------------------------------")
# dictionary creation
db={7:"salem",1:"chennai",3:"trichy",4:"mumbai",19:"london",21:"newyork"}
print("Dictionary Contents, Before Deletion")
print(db)
print("-----------------------------------------------")
# delete the key 19 using del keyword
del db[19]
```

Python Dictionary Creation

```
# delete the key 21 using pop()
db.pop(21)
print("Dictionary Contents, After Deletion")
print(db)
print("----------------------------------------------")
# deletion of all elements from dictionary
db.clear()
print("Dictionary Contents, After Clear()")
print(db)
print("----------------------------------------------")
# deletion of entire dictionary
del db
print("Dictionary Contents, After Complete Deletion")
print(db)
print("----------------------------------------------")
```

2. OUTPUT



V. DICTIONARY UPDATION

(Storage, Updation and Retrieval)

(welcome.py)

Language : Python 3.7 (Python 3)

Application Type : Console App

1. SOURCE CODE

```
print("----------------------------------------")
print("\tPython Dictionary")
print("----------------------------------------")
# dictionary creation
ele=dict({5:"Salem",7:"Chennai",4:"Madurai",1:"Trichy"})
print("Before Updation, Dictionary Contents")
print(ele)
print("----------------------------------------")
# dictionary updation (updating two keys)
ele[7]="Nellai"
ele[1]="Kovai"
print("After Updation, Dictionary Contents")
print(ele)
print("----------------------------------------")
```

Python Dictionary Creation

Python Dictionary Updation using keys

2. OUTPUT

```
C:\WINDOWS\system32\cmd.exe
-----------------------------------------
        Python Dictionary
-----------------------------------------
Before Updation, Dictionary Contents
{5: 'Salem', 7: 'Chennai', 4: 'Madurai', 1: 'Trichy'}
-----------------------------------------
After Updation, Dictionary Contents
{5: 'Salem', 7: 'Nellai', 4: 'Madurai', 1: 'Kovai'}
-----------------------------------------

C:\Users\Ganesh\Desktop>
```

VI. INDIVIDUAL ITEM ACCESS OF DICTIONARY

(Storage, and Retrieval)

(welcome.py)

Language : Python 3.7 (Python 3)

Application Type : Console App

Python Dictionary Creation

1. SOURCE CODE

```
print("-------------------------------------------------------------")
print("\tIndividual Item of Dictionary")
print("-------------------------------------------------------------")
# dictionary creation
ele={5:"Salem",7:"Chennai",4:"Madurai",1:"Trichy"}
print(ele)
print("-------------------------------------------------------------")
# accessing individual items of dictionary
r1=ele.get(7)
r2=ele[4]
print("Element at key 7 \t: ",r1)
print("Element at key 4 \t: ",r2)
print("-------------------------------------------------------------")
```

2. OUTPUT

```
C:\WINDOWS\system32\cmd.exe

C:\Users\Ganesh\Desktop>python welcome.py
------------------------------------------------------------
        Individual Item of Dictionary
------------------------------------------------------------
{5: 'Salem', 7: 'Chennai', 4: 'Madurai', 1: 'Trichy'}
------------------------------------------------------------
Element at key 7        :  Chennai
Element at key 4        :  Madurai
------------------------------------------------------------

C:\Users\Ganesh\Desktop>_
```